



To Churn or Not to Churn? Modelling the Factors Behind Credit Card Churn

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Introduction

A popular advertising tactic involves offering welcome bonuses to lure customers to a business. While effective at expanding the customer base, many exploit these incentives, signing up just for the perks and then discontinuing their patronage. This challenge is especially notable for banks offering such benefits to attract customers to their credit card programs, resulting in losses from underutilized credit cards. Hence, banks must recognize signs of *churn* (business slang for customers who cease service) to filter out future opportunistic customers.) to filter out future opportunistic customers.

The study's goal is to discover which characteristics can help predict a customer's likelihood to churn and to find any correlation between any of the factors.

Number of Products, Income, and Churn

Let's look at the correlation between the number of bank products owned (ie. an account or credit card counting as one product), customer income, and churn status by grouping our data by these income groups and calculating the mean number of products and proportion of customers who churned of each income group.

Income_Category	Mean Product Count	Churn Proportion
\$120K +	3.77854	0.173315
\$40K - \$60K	3.78324	0.151397
\$60K - \$80K	3.83381	0.134807
\$80K - \$120K	3.81694	0.157655
Less than \$40K	3.82336	0.171862

Figure 1. A table of the customers grouped by income category along with each category's respective mean number of products and churn proportion.

From Figure 1, we can see that the \$60K-80K income group has the lowest churn proportion. To determine whether this observation is statistically significant, we will run hypothesis testing based on the null hypothesis that the lower churn proportion is purely due to random chance and the alternative hypothesis that the difference occurs for reasons outside of random chance.

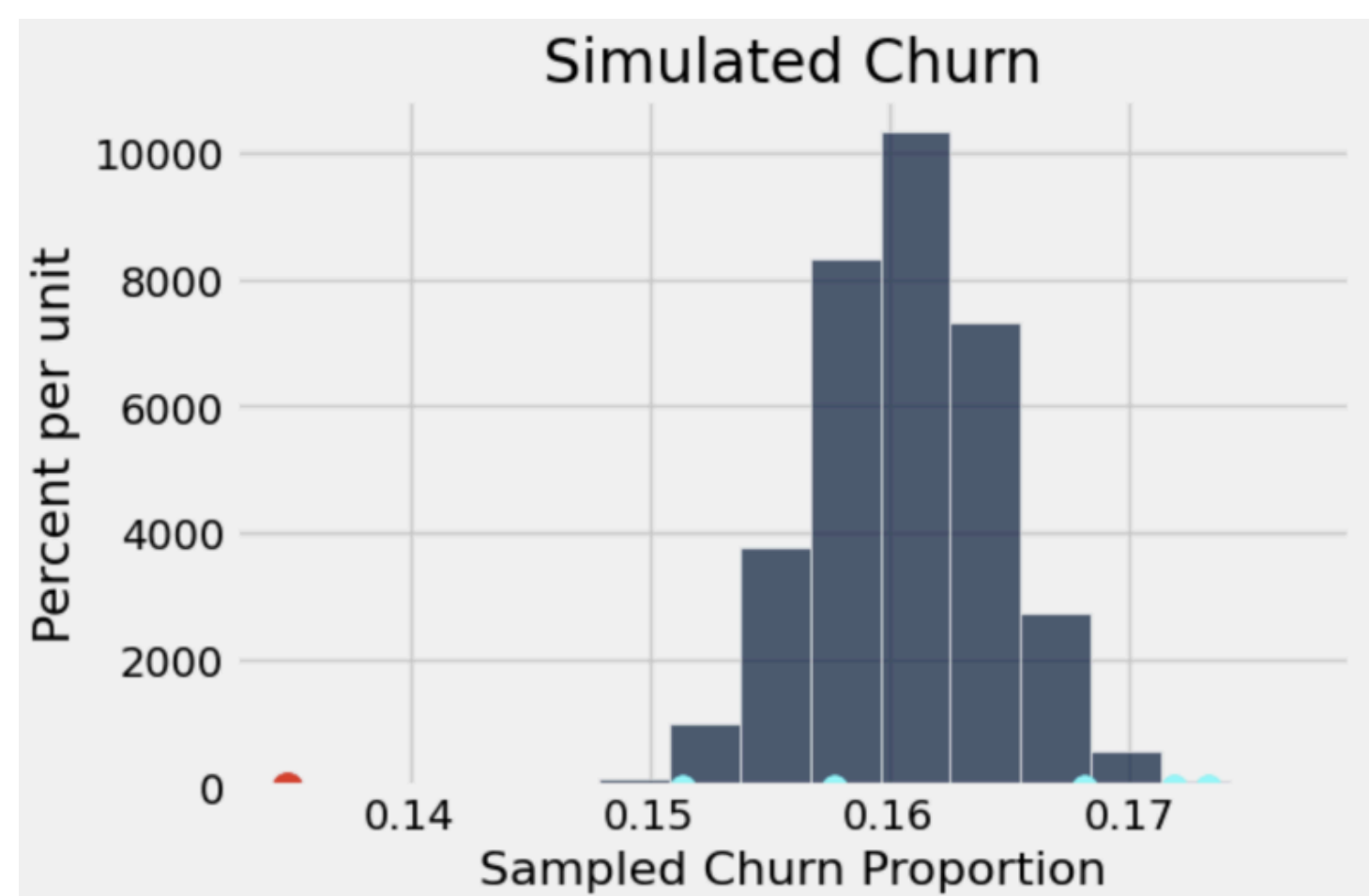


Figure 2. A histogram containing the results of simulation based on a churn proportion of 0.16065962 amongst 10127 customers. The red dot shows our observed value of a 0.134807 churn proportion amongst customers who make \$60K-\$80K.

From Figure 5, we can see that having a churn proportion of 0.134807 is out of the ordinary for a population that has an average proportion of 0.16065962. The p-value of this test can be visually discerned as 0, supporting our alternative hypothesis that the lower churn proportion in the \$60-80K income group is statistically significant and not a product of random chance.

Average Revolving Balance

Since churned customers are aiming to cancel their cards after redeeming benefits, we'd expect them to accumulate less credit card debt, which is measured using the average revolving balance variable (the portion of unpaid credit card spending at the end of a billing cycle).

After running simulations to estimate the proportion of churners who have a lower-than-average revolving balance we obtain the results shown below in Figure 3.

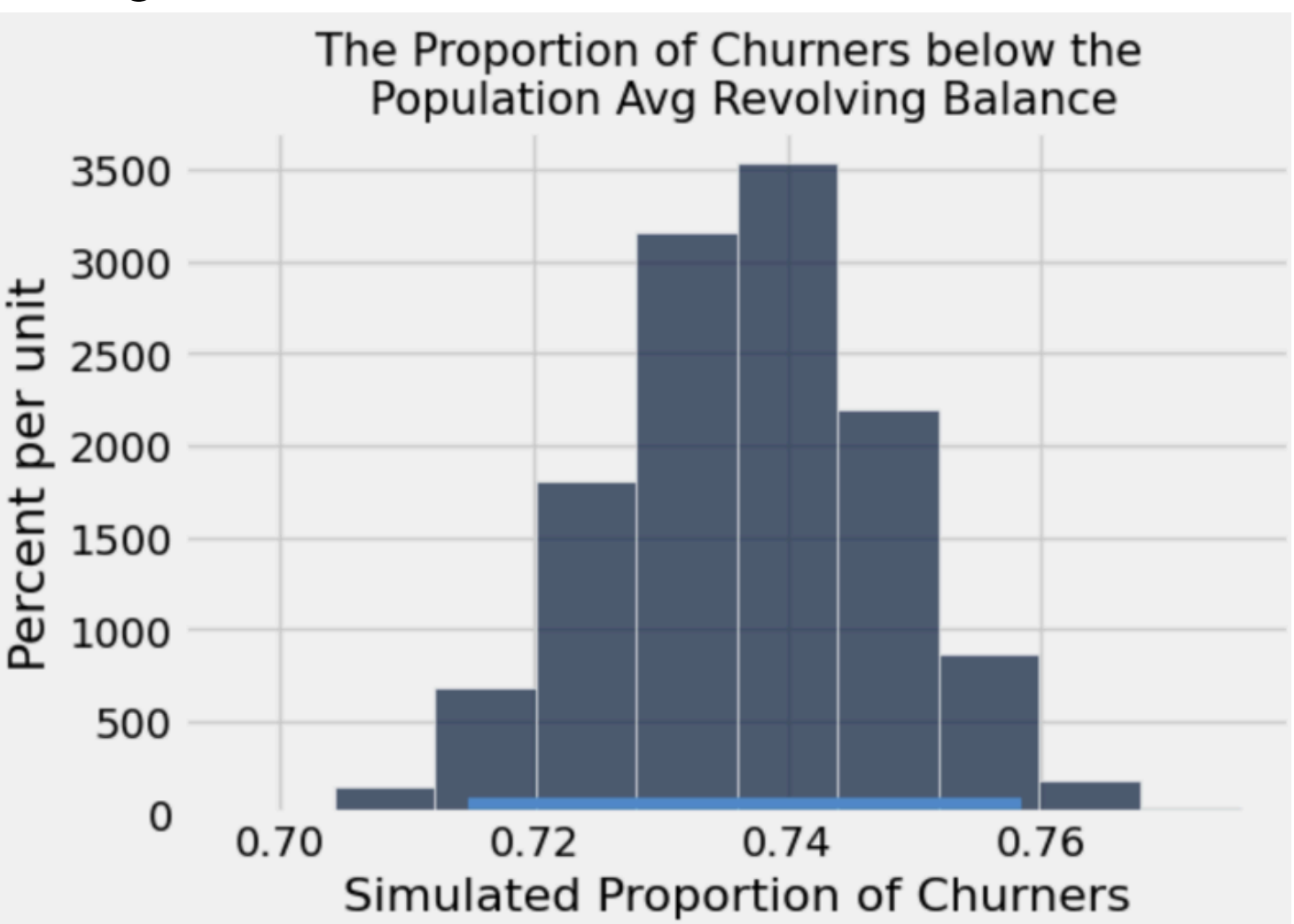


Figure 3. The results of 10,000 simulations of the proportion of churned customers who lie below the population average for average revolving balance generated using bootstrapping.

From Figure 3, we have a 95% confidence interval of (0.725, 0.758) shown in light blue. This confirms that a majority of churned customers have lower than average values for average revolving balance.

Regression Analysis and Modelling

Knowing that the average revolving balance is an indicator of churn, we can construct a linear model using similar variables, average utilization ratio (the percentage of credit used) and credit limit (the spending cap on a credit card).

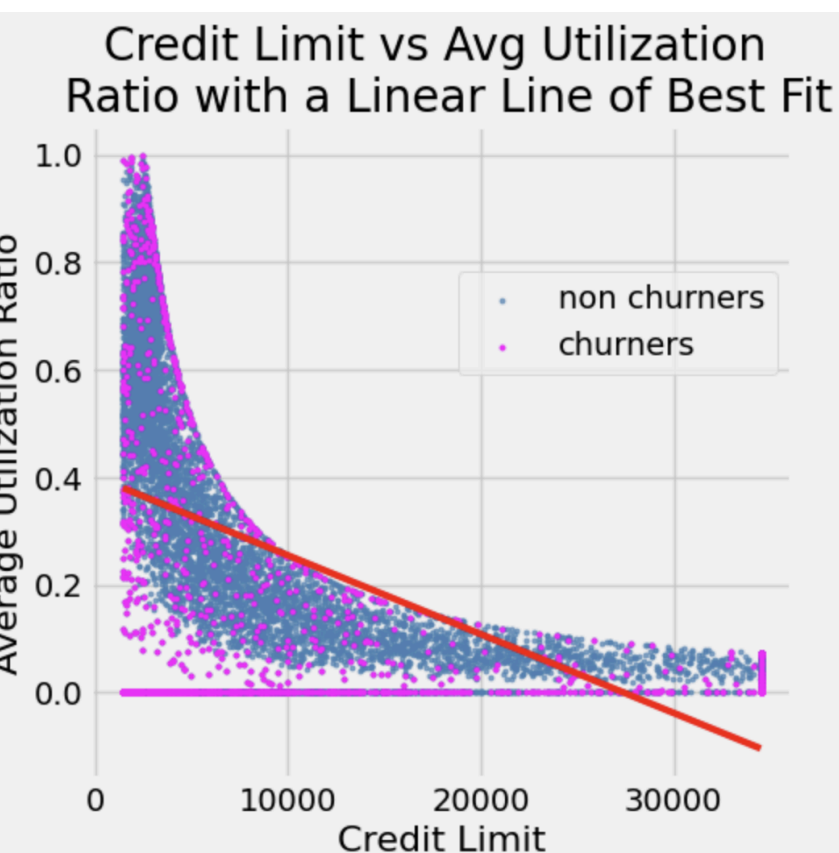


Figure 4. A plot of the credit limit and the average utilization ratio for our dataset with a linear line of best fit plotted. Each point is color-coded to indicate the customer's churn status with pink for those who churned and blue for those who did not.

From Figure 4, we see that the two variables have a logarithmic relationship and are negatively correlated.

Seeing the logarithmic relation, we can improve our model by plotting them in a log-log scale to linearize our data and find a more accurate line of best fit. The results of the linearization are shown below in Figures 5 and 6.

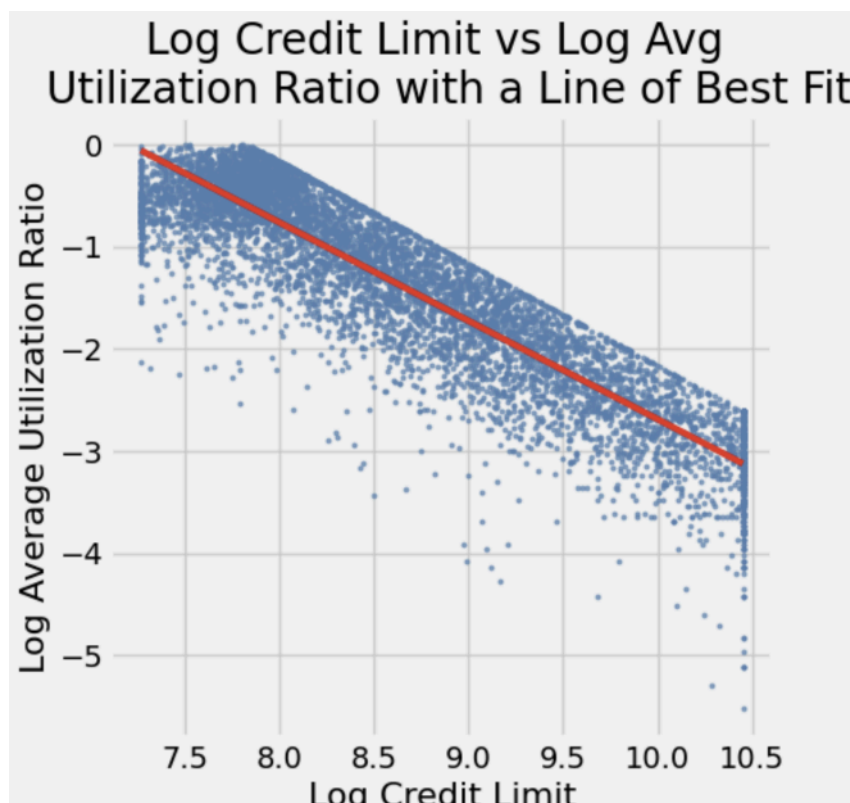


Figure 5. The graph of the log credit limit and the log average utilization ratio for the 10127 customers in our data set with a line of best fit plotted.

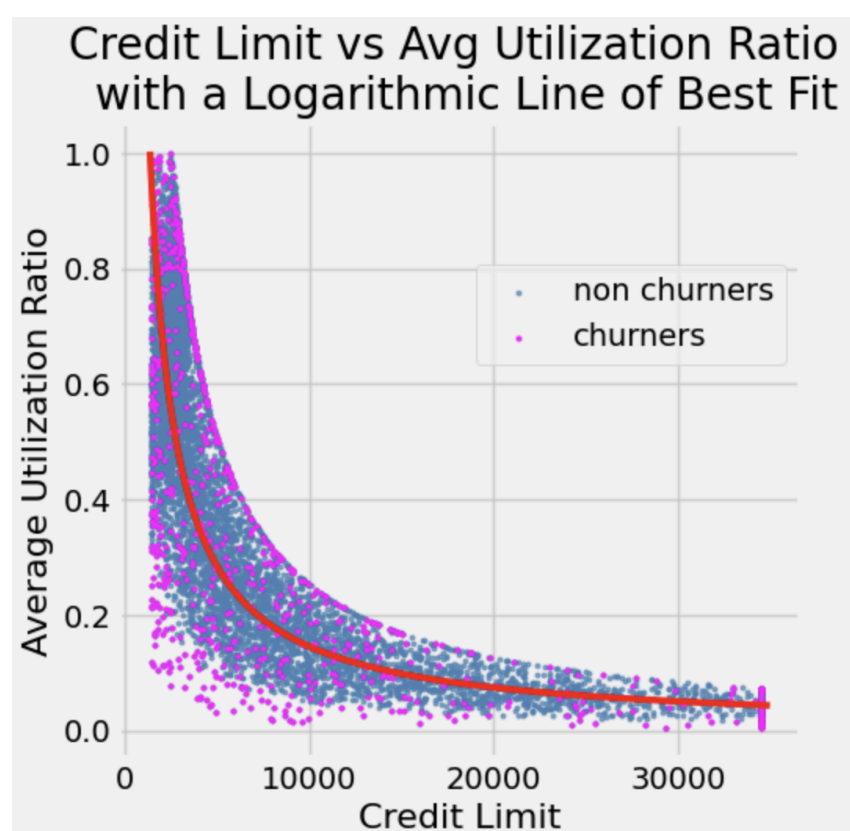


Figure 6. The graph of credit limit and average utilization ratio with a logarithmic line of best fit.

Now that we have a fairly accurate model, we can use the pink "churn region" that lies below the logarithmic grouping of non-churning customers to predict the churn of future customers.

Citations

Goyal Sakashi. *Credit Card customers*. (2020, November 19). Kaggle. <https://www.kaggle.com/datasets/sakshigoyal17/credit-card-customers/data>
MoneyGeek Team. (2024, March 21). *Credit card glossary: Basic terms to know*. MoneyGeek.com. <https://www.moneygeek.com/credit-cards/resources/credit-card-glossary/>